

Teacher: CVEK FINS KERJ MAYR SCHP TRAK XIAR

NAME: _____

Mathematical Methods Unit 3

Test 4A Exponentials and Logs

Tech - Free

Question and Answer Book

VCE Test

- Reading time is **5 minutes**
- Writing time is **50 minutes**

Materials supplied

- Question and Answer Book of 8 pages
- Formula Sheet

Students are **not** permitted to bring any technology (calculators or software) or notes of any kind into the examination room.

Students are **not** permitted to bring mobile phones and/or any unauthorised electronic devices into the examination room.

Contents

5 questions (33 marks)

pages
2-8

Instructions

- Answer **all** questions in the spaces provided.
 - Write your responses in English.
 - Unless otherwise specified, an **exact** answer is required for each question.
 - In questions where more than one mark is available, appropriate working **must** be shown.
 - Unless otherwise indicated, the diagrams in this book are **not** drawn to scale.
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Question 1 (4 marks)

Fully simplify the following expressions expressing your answers as a single term:

a. $\frac{4^{2a-b} 8^{b-3a}}{(16^{a+b})^{-2}}$

2 marks

b. $6\log_2(\sqrt{x}) - 2\log_2(2x) + 3, x > 0$

2 marks

Question 2 (6 marks)Solve the following equations for x :

a. $2\log_e(x+2) - \log_e(x) = \log_e(2x+1).$

3 marks

b. $3e^x = 5 + e^{-x}.$

3 marks

Question 3 (10 marks)

Consider the functions f and g with rules $f(x) = e^{3x}$ and $g(x) = \log_e(x - 2)$ defined on their maximal domains.

- a.** State the domain and range of f and g . 2 marks

- b.** Show that the composite function $f \circ g(x)$ is defined. 1 mark

- c.** Find the rule for the composite function $f \circ g(x)$ in the form $f \circ g(x) = (ax + b)^n$, $a, b, n \in \mathbb{Z}$. 2 marks

- d. Define a suitable restriction of the domain of f so that the function $g \circ f(x)$ is defined on its maximal domain.

3 marks

- e. Find the function $g \circ f(x)$, expressing your answer in full function notation.

2 marks

Question 4 (3 marks)

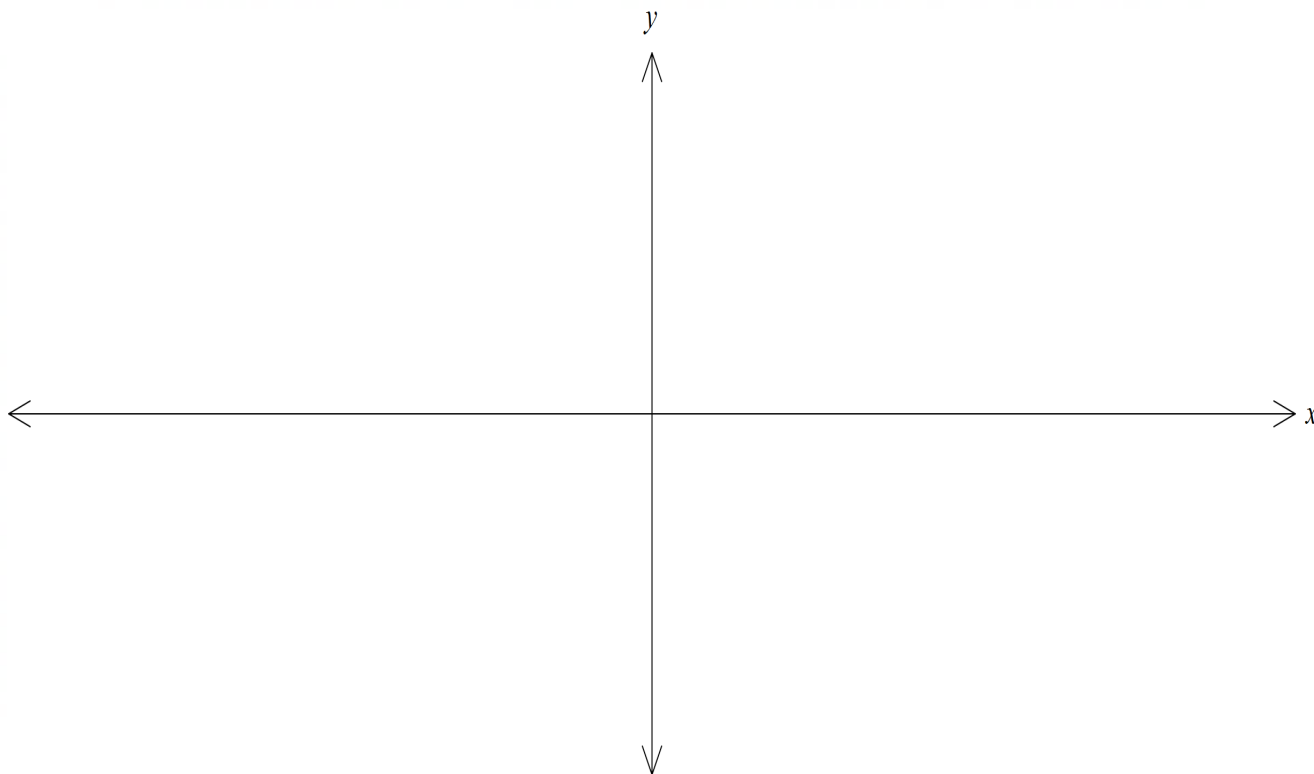
Consider the function $h: R \rightarrow R$, $h(x) = e^{3x}$. Find the value(s) of k , where $k \in R^+$, such that the tangent to the curve $y = h(x)$ at the point $x = k$ will have a positive y -intercept.

3 marks

Question 5 (10 marks)

Consider the function $k : (-\frac{1}{3}, \infty) \rightarrow \mathbb{R}$, $k(x) = 2 \log_e(3x+1) - 1$.

- a.** Sketch the graph of $y = k(x)$ on the axes below labelling intercepts with their coordinates and asymptotes with their equations.

3 marks

- b.** Find the rule for the inverse function $k^{-1}(x)$.

2 marks

- c.** Sketch the graph of the inverse function $y = k^{-1}(x)$ on the set of axes in part **a**, labelling intercepts with their coordinates and asymptotes with their equations.

2 marks

- d.** Find the coordinates of the point on the graph of $y = k(x)$, where the gradient of the tangent to to the curve $y = k(x)$ is equal to 1.

3 marks

End of test questions

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